

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1521

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I D.G.Raghavi (192521173) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student: D.G.Raghavi (192521173)

Assessment Tasks Short Answer Test

1. **Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.**

Definition

Cloud computing is a technology that provides computing services such as servers, storage, databases, networking, and software over the internet. Users can access these resources whenever needed without purchasing and maintaining their own physical hardware.

For example, Google Drive allows users to store files online and access them from different devices without maintaining a personal storage server.

Four characteristics of cloud computing

1. On-demand self-service

Users can access computing resources such as storage, virtual machines, and databases whenever they need them. They can create or remove resources without contacting the service provider.

Example: A developer can create a virtual server on AWS within minutes.

2. Broad network access

Cloud services are available through the internet and can be accessed using laptops, computers, tablets, and smartphones. Users can access their data from different locations.

Example: A student can open files stored in Google Drive using either a mobile phone or a laptop.

3. Resource pooling

Cloud providers maintain large groups of computing resources and share them among multiple users. Resources are allocated according to each user's requirements, improving hardware utilization and reducing costs.

Example: Multiple companies can use the same cloud provider's infrastructure while keeping their data logically separated.

4. Rapid elasticity

Cloud resources can be increased or decreased quickly depending on demand. This allows organizations to handle sudden increases or decreases in workload.

Example: An online shopping website can increase its server capacity during festival sales and reduce it afterward.

Difference between cloud computing and traditional on-premise computing

Cloud computing	Traditional on-premise computing
Resources are accessed over the internet.	Resources are maintained within the organization.
Requires less initial hardware investment.	Requires purchasing physical servers and equipment.
Resources can be scaled quickly.	Scaling often requires buying and installing hardware.
The cloud provider manages the physical infrastructure.	The organization manages its own physical infrastructure.

Conclusion: Cloud computing provides flexible, scalable, and cost-effective services, whereas traditional computing requires organizations to purchase and maintain their own infrastructure.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

Cloud computing developed through several stages of technological advancement. Improvements in hardware, networking, virtualization, and internet-based software made modern cloud services possible.

Stages in the evolution of cloud computing

1. Mainframe computing

In the early stages, organizations used large and expensive mainframe computers. Multiple users accessed a central computer through connected terminals. This introduced the idea of sharing computing resources.

2. Personal computer evolution

Personal computers became smaller, more affordable, and widely available. Users could perform computing tasks locally, reducing dependence on centralized computers.

3. Client-server computing

In this model, client computers request services from centralized servers through a network. Servers store data, process requests, and provide information to clients.

Example: A web browser requests a webpage from a web server.

4. Distributed computing

Multiple computers connected through a network began working together to share processing power, storage, and data. This improved performance and supported larger workloads.

5. Virtualization

Virtualization allowed a single physical server to run multiple virtual machines. Each virtual machine could have its own operating system and applications. This improved resource utilization and reduced hardware costs.

6. Internet and web-based software evolution

The growth of the internet and web browsers made it possible to access applications remotely. Users no longer needed to install every application directly on their computers.

Example: Gmail allows users to send and receive emails through a web browser.

7. Modern cloud computing

Cloud providers combined powerful servers, virtualization, distributed systems, and internet technologies to deliver computing resources on demand.

Examples include Amazon Web Services (AWS), Microsoft Azure, and Google Cloud.

How these developments enabled modern cloud platforms

- Powerful hardware provides the computing capacity.
- Virtualization enables flexible allocation of resources.
- Distributed systems support large-scale workloads.
- Internet technologies allow remote access.
- Web-based software enables services to be delivered online.

Conclusion: The evolution from mainframes to virtualized servers and internet-based applications created the foundation for modern cloud computing. Today, cloud platforms provide scalable resources and software without requiring users to maintain their own physical infrastructure.

3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Server virtualization is a technology that divides one physical server into multiple virtual machines. Each virtual machine can run its own operating system and applications.

Role of server virtualization:

- Improves hardware utilization.
- Reduces hardware and maintenance costs.
- Allows multiple operating systems to run on one server.
- Enables flexible resource allocation.
- Supports rapid creation and management of virtual machines.

Difference:

Virtualization	Cloud Computing
A technology that creates virtual resources.	A service model that delivers computing resources over the internet.
Can operate within a local data center.	Commonly provides remote, on-demand services.
Focuses on resource abstraction and sharing.	Focuses on service delivery, scalability, and metered usage.
Example: VMware ESXi.	Example: Amazon Web Services (AWS).

Conclusion: Virtualization is an important enabling technology for cloud computing, while cloud computing uses technologies such as virtualization to deliver services to users.

4. Discuss the role of data centers in delivering cloud services..

Definition

A data center is a physical facility that contains servers, storage devices, networking equipment, power systems, and cooling systems. It provides the infrastructure required to store, process, and manage digital information.

Cloud providers operate data centers to deliver services to individuals, businesses, and organizations.

Role of data centers in cloud computing

1. Data storage

Data centers contain storage systems that store files, databases, backups, and application information. Users can access their data whenever required.

Example: Google Drive stores user files using cloud infrastructure.

2. Data processing

Servers in data centers process user requests, execute applications, and perform calculations. They provide the computing power required to run cloud services.

Example: When a user opens an online application, servers process the request and return the required information.

3. Network connectivity

Routers, switches, and other networking equipment connect servers and storage systems. They also enable communication between data centers and users over the internet.

4. Security

Data centers use physical access controls, firewalls, monitoring systems, encryption, and other security measures to help protect information from unauthorized access.

5. High availability and reliability

Cloud providers use redundant servers, backup power, and alternative network connections to reduce service interruptions. Some services also replicate data across multiple locations to support disaster recovery.

6. Scalability

Data centers allow cloud providers to add servers, storage, and networking capacity as demand increases. This enables organizations to handle growing workloads without purchasing their own physical infrastructure.

7. Resource management

Cloud providers monitor computing resources and allocate them to customers according to demand. Automation helps manage workloads and improve infrastructure utilization.

8. Backup and disaster recovery

Data centers can maintain backup copies and recovery systems to help restore data and services after hardware failures or other disruptions.

Real-world example

When a student uploads an assignment to Google Drive:

1. The file is transmitted through the internet.
2. Cloud infrastructure receives and stores the file.

3. Storage systems retain the file.
4. Networking systems allow the student to access it later from another device.
5. Backup and security mechanisms help protect the data.

Conclusion: Data centers form the physical foundation of cloud computing. They provide the storage, processing, networking, security, and reliability needed to deliver cloud services efficiently.

5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Cloud computing provides different service models according to the level of infrastructure, software, and management offered to users. The major models are IaaS, PaaS, SaaS, and XaaS.

1. IaaS – Infrastructure as a Service

IaaS provides virtual servers, storage, networking, and other basic computing resources over the internet. Users can install operating systems and applications on the provided infrastructure.

- Provider responsibility: Physical servers, networking, storage hardware, and virtualization infrastructure.
- User responsibility: Operating system, applications, data, and many configuration and security tasks.
- Example: Amazon EC2, which provides virtual servers for running applications.

Use: Suitable for organizations that need control over their computing environment without purchasing physical servers.

2. PaaS – Platform as a Service

PaaS provides a ready-made environment for developing, testing, and deploying applications. Developers can focus on writing code without managing the underlying physical infrastructure and much of the platform software.

- Provider responsibility: Infrastructure, operating system, runtime environment, and platform maintenance.
- User responsibility: Application code, application data, and application-level configuration and security.
- Example: Google App Engine, which provides a platform for deploying applications.

Use: Suitable for developers who want to build applications without managing servers and operating systems directly.

3. SaaS – Software as a Service

SaaS provides ready-to-use software over the internet. Users can access the application through a web browser or app without installing and maintaining the underlying server infrastructure.

- Provider responsibility: Application hosting, infrastructure, software maintenance, and updates.
- User responsibility: Account management, data input, access permissions, and application settings.
- Example: Gmail, which allows users to send and receive emails online.

Use: Suitable for users who need software without managing its installation or infrastructure.

4. XaaS – Everything as a Service

XaaS is a broad term describing different IT resources and capabilities delivered as services over a network. It can include storage, security, networking, software, and computing resources.

- Provider responsibility: Depends on the service; it may include infrastructure, service operation, maintenance, and support.
- User responsibility: Selecting suitable services, managing access and configuration, and using resources securely.
- Example: Microsoft 365, which provides cloud-based productivity applications and related services.

Use: Suitable for organizations that want to obtain multiple IT capabilities as services instead of managing everything themselves.

Comparison of cloud service models

Feature	IaaS	PaaS	SaaS	XaaS
Main offering	Infrastructure	Development platform	Ready-to-use software	Broad range of IT services
User focus	Managing systems and applications	Developing applications	Using software	Selecting and using services
Provider manages	Physical infrastructure	Infrastructure and platform	Application and infrastructure	Depends on the service
Example	Amazon EC2	Google App Engine	Gmail	Microsoft 365

Conclusion

IaaS provides computing infrastructure, PaaS provides an application development platform, SaaS delivers ready-to-use software, and XaaS covers a broad range of services delivered over a network. These models allow users to select services according to their needs while reducing the burden of managing physical infrastructure.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected subpoints completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow